

Project Management Plan & Estimation

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Project Management Plan & Estimation Specification

Project WikiMove – Migrating a Company Knowledge Base to a New Platform

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1. Executive Summary & Project Context

Project WikiMove is an enterprise-scale software engineering documentation and migration initiative designed to decommission an aging, ungoverned internal wiki comprising 14,000 pages accumulated over 11 years. The organizational imperative is driven by an immovable external constraint: the legacy platform software licence expires in 9 months (~39 weeks).

The initiative is led by a dedicated Technical Writer operating on a part-time project allocation of 3 person-days per week, supported by 12 engineering teams contributing 1 person-day per week each. The core challenge is addressing the mathematical schedule conflict between total required effort (630 person-days = 42.0 weeks) and the 39-week licence window, while instituting sustainable content ownership to prevent future wiki decay.

2. Project Objectives & Boundary Management

2.1 Project Objectives

- OBJ-01: Complete 100% triage of all 14,000 legacy pages into Keep, Archive, or Delete.
- OBJ-02: Execute high-fidelity migration of exactly 7,000 retained pages (50% assumption) to the new platform.
- OBJ-03: Assign 100% of migrated content to validated named human custodians (Zero Orphan Policy).
- OBJ-04: Achieve 0.0% broken internal links (NFR-01) and sub-second search discoverability (NFR-02).
- OBJ-05: Formulate and manage contingency options for the 3-week schedule gap.

2.2 Boundary Management (Academic Scope)

In accordance with academic instructions, WikiMove is an academic software engineering and project management proposed system. Project deliverables focus on complete technical specifications, rigorous mathematical estimation,

WBS, Gantt scheduling, risk frameworks, and static UI mockups. No backend services, cloud infrastructures, databases, or production APIs are deployed.

3. Project Phases & Lifecycle Model

WikiMove adopts a phased linear-iterative lifecycle structured into 9 coherent academic phases:

```
graph TD
    P1[Phase 1: Project Initiation] --> P2[Phase 2: Requirements & Architecture]
    P2 --> P3[Phase 3: Wiki Inventory & Cataloging]
    P3 --> P4[Phase 4: Content Triage 40 p/d]
    P4 --> P5[Phase 5: Migration Preparation]
    P5 --> P6[Phase 6: Content Migration 25 p/d]
    P6 --> P7[Phase 7: Testing & QA Validation]
    P7 --> P8[Phase 8: Ownership & Governance]
    P8 --> P9[Phase 9: Project Closure]
```

- Phase 1 — Initiation (Weeks 1–2): Project charter approval, stakeholder alignment, governance setup.
- Phase 2 — Requirements & Architecture (Weeks 3–4): IEEE 830 SRS, BRD baseline, UML modeling, UI prototyping.
- Phase 3 — Wiki Inventory (Weeks 5–6): Automated metadata export of 14,000 pages, team domain mapping.
- Phase 4 — Content Triage (Weeks 7–22): 14,000 pages triaged at 40 pages/day across 12 teams + TW.
- Phase 5 — Migration Preparation (Weeks 20–22): URL rewrite registry, formatting templates, target schema setup.
- Phase 6 — Content Migration (Weeks 23–36): 7,000 kept pages migrated in sprint batches at 25 pages/day.
- Phase 7 — Testing & QA Validation (Weeks 34–38): Link integrity scanning, search indexing QA, defect resolution.
- Phase 8 — Ownership & Governance (Weeks 37–40): Named owner binding, 90-day review rules, handover.
- Phase 9 — Project Closure (Weeks 41–42): Final audit, closure note, lessons learned, legacy shutdown.

4. Work Breakdown Structure (WBS)

```
1.0 Project WikiMove
% % % 1.1 Project Initiation & Governance
% % % 1.1.1 Project Charter & Stakeholder Matrix
% % % 1.1.2 Team Allocation & Cadence Setup (15 p-d/wk)
% % % 1.2 Requirements & Architecture
% % % 1.2.1 Business Requirements Document (BRD)
% % % 1.2.2 IEEE 830 Software Requirements Specification (SRS)
% % % 1.2.3 UML Design Package (7 Diagrams)
% % % 1.2.4 Static UI Prototype Design (6 Screens)
% % % 1.3 Wiki Inventory & Ingestion
% % % 1.3.1 14,000 Page Legacy Extraction
% % % 1.3.2 Domain Tagging & 12 Team Partitioning
% % % 1.4 Content Triage (14,000 Pages @ 40 p/d = 350 p-d)
% % % 1.4.1 Engineering Domain Review (Keep / Archive / Delete)
% % % 1.4.2 Orphan Page Reclamation by Technical Writer
% % % 1.4.3 Archival Cold Storage Packaging (3,500 Pages)
% % % 1.5 Migration Planning & Tooling Setup
% % % 1.5.1 URL Rewrite Engine & 301 Redirection Schema
% % % 1.5.2 Migration Batch Construction (7,000 Kept Pages)
% % % 1.6 Content Migration (7,000 Kept Pages @ 25 p/d = 280 p-d)
% % % 1.6.1 Batch Execution (Sprint Batches 1..10)
% % % 1.6.2 Asset & Code Snippet Formatting Transposition
% % % 1.7 Testing, Verification & Quality Assurance
% % % 1.7.1 Automated Link Integrity Crawler (NFR-01)
% % % 1.7.2 Search Index & Ranking Validation (NFR-02)
% % % 1.7.3 DRE & Defect Density Metric Logging
% % % 1.8 Ownership & Post-Migration Governance
% % % 1.8.1 Named Owner Binding (Zero Orphan Policy)
% % % 1.8.2 90-Day Periodic Review Workflow Configuration
% % % 1.9 Project Closure & Handover
% % % 1.9.1 Final Project Report & Academic Presentation
% % % 1.9.2 Decommission Audit & Licence Handover
% % % 1.9.3 Lessons Learned & Post-Mortem Documentation
```

5. Activity Precedence, Duration & Dependencies

Task ID	Activity Description	Predecessor	Duration (Weeks)	Required Effort (p-d)	Primary Resource
A1	Project Initiation & Chartering	None	2	10	PM, CTO, TW
A2	Requirements Engineering (BRD & SRS)	A1	2	20	BA, TW
A3	Architecture, UML & UI Mockups	A2	2	15	System Architect, UI Designer
A4	Legacy 14k Page Inventory Ingestion	A1	2	10	TW, Data Analyst
A5	Content Triage (14,000 pages @ 40 p/d)	A3, A4	16	350	12 Teams + TW (15 p-d/wk)
A6	Migration Prep & URL Rewrite Rules	A4	3	15	TW, QA Engineer
A7	Migration Execution (7k pages @ 25 p/d)	A5, A6	14	280	12 Teams + TW (15 p-d/wk)
A8	Link & Search QA Testing	A7	4	20	QA Engineer, TW
A9	Ownership Binding & Governance Rules	A5, A7	3	15	TW, Eng Leads
A10	Final Decommission & Project Closure	A8, A9	2	10	PM, CTO, TW

6. Milestone Schedule

Milestone ID	Milestone Deliverable / Event	Scheduled Week	Critical Status
M1	Project Charter & Baseline Requirements Approved	Week 4	Baseline Milestone
M2	14,000 Legacy Inventory Extracted & Partitioned	Week 6	Input to Triage
M3	Mid-Point Triage Review (7,000 Pages Triaged)	Week 14	Schedule Tracking Gate
M4	100% Triage Complete (14,000 Pages Processed)	Week 22	Triage Exit Gate
M5	First 2,500 Kept Pages Migrated & Verified	Week 28	Migration Velocity Check
M6	100% Kept Pages Migrated (7,000 Pages on Target)	Week 36	Core Technical Goal
M7	Legacy Platform Software Licence Expiration	Week 39	HARD DEADLINE (39 Wks)
M8	Final QA Sign-off & Governance Certification	Week 40	Quality Acceptance
M9	Full Project Closure (Baseline Completion)	Week 42	3-Week Overrun Beyond M7

7. Resource Allocation & Weekly Capacity Modeling

The project operates under

a strict, non-negotiable weekly resource ceiling:

$$\text{Technical Writer Allocation} = 3 \text{ person-days / week}$$

$$\text{12 Engineering Teams Allocation} = 12 \times 1 \text{ person-day / week} = 12 \text{ person-days / week}$$

$$\text{Total Weekly Capacity } (C) = 3 + 12 = 15 \text{ person-days / week}$$

This fixed capacity ceiling dictates that any task must be scheduled within 15 person-days per calendar week unless formal management contingencies are triggered.

8. Mathematical Estimation Sheet (Detailed Formulas)

The estimation model is derived strictly from the case-study empirical data:

```
graph TD
    A["Total Pages = 14,000"] --> B["Outdated = 50% (7,000)"]
    B --> C["Kept Pages = 50% (7,000)"]
    C --> D["Triage Effort: 14,000 / 40 = 350 Person-Days"]
    D --> E["Migration Effort: 7,000 / 25 = 280 Person-Days"]
    E --> F["Total Project Effort: 350 + 280 = 630 Person-Days"]
    F --> G["Weekly Capacity: 3 + 12 = 15 Person-Days/Week"]
    G --> H["Estimated Duration: 630 / 15 = 42.0 Weeks"]
    H --> I["Licence Period: 9 Months = 39.0 Weeks"]
    I --> J["Schedule Gap: 42.0 - 39.0 = +3.0 Weeks"]
```

Overrun"]
H --> J

Step 1: Inventory & Retention Volume

$$\begin{aligned} & \text{Total Wiki Inventory } (N) = 14,000 \text{ pages} \\ & \text{Outdated Ratio } (r_{\text{outdated}}) = 50\% \\ & \text{implies } 14,000 \times 0.50 = 7,000 \text{ pages} \\ & \text{Kept Pages } (N_{\text{kept}}) = 14,000 \times 50\% = \\ & \mathbf{7,000 \text{ pages}} \end{aligned}$$

Step 2: Triage Effort Calculation

$$\begin{aligned} & \text{Triage Velocity } (v_{\text{triage}}) = 40 \text{ pages / person-day} \\ & \text{Triage Effort } (E_{\text{triage}}) = \frac{N}{v_{\text{triage}}} = \frac{14,000}{40} = \\ & \mathbf{350 \text{ person-days}} \end{aligned}$$

Step 3: Migration Effort Calculation

$$\begin{aligned} & \text{Migration Velocity } (v_{\text{migration}}) = 25 \text{ pages / person-day} \\ & \text{Migration Effort } (E_{\text{migration}}) = \frac{N_{\text{kept}}}{v_{\text{migration}}} = \frac{7,000}{25} = \\ & \mathbf{280 \text{ person-days}} \end{aligned}$$

Step 4: Total Project Effort

$$\begin{aligned} E_{\text{total}} &= E_{\text{triage}} + E_{\text{migration}} = 350 + 280 \\ &= \mathbf{630 \text{ person-days}} \end{aligned}$$

Step 5: Estimated Project Duration

$$\begin{aligned} D &= \frac{E_{\text{total}}}{C} = \frac{630 \text{ person-days}}{15 \text{ person-days / day}} \end{aligned}$$

$$\text{week}} = \mathbf{42.0 \text{ weeks}}$$

Step 6: Licence Expiry Timeline & Gap Calculation

$$\text{Licence Period } (T_{\text{licence}}) = 9 \text{ months}$$

$$\text{Standard Academic Conversion Rate} = 4.333 \text{ weeks / month}$$

$$T_{\text{licence}} = 9 \times 4.333 \approx \mathbf{39.0 \text{ weeks}}$$

$$\text{Schedule Gap } (\Delta T) = D - T_{\text{licence}} = 42.0 - 39.0 = \mathbf{+3.0 \text{ weeks}}$$

(Schedule Overrun))

9. Gantt Chart Representation (42 Weeks vs. 39 Weeks)

The Gantt schedule below illustrates the 9 phases across 42 weeks, highlighting the 39-week licence expiration boundary and the 3-week conflict.

Activity / Phase				Wk
1-4	Wk	5-8	Wk	
9-12	Wk	13-16	Wk	
17-20	Wk	21-24	Wk	
25-28	Wk	29-32	Wk	
33-36	Wk	37-39	[Wk	
39: LICENCE]	Wk	40-42		

Phase 1: Initiation				[====]
Phase 2: Requirements & Arch				[====]
Phase 3: Wiki Inventory				[====]
Phase 4: Content Triage (350 p-d)				[=====]
				=====
				=====]

- Critical Activities (Zero Slack):
 - Activity A5 (Content Triage, 350 person-days): 16 weeks duration on critical path.
 - Activity A7 (Content Migration, 280 person-days): 14 weeks duration on critical path.
 - Total Critical Path Duration: Exactly 42.0 weeks, exceeding the 39.0-week licence limit.

Because the baseline estimate exceeds the licence expiration by 3.0 weeks, the project manager must implement academic contingency strategies:

2. Strategy 2 — Resource Uplift (Capacity Compression): Increase the Technical Writer's allocation from 3 to 5 person-days/week during Weeks 15–35 (adding +40 person-days) or allocate 1 additional engineer per team during final migration, compressing duration from 42 weeks to 37.5 weeks.

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